

Relational schema mapping

Tables

```
DEPARTMENT(  
  department_id PK,  
  name UNIQUE,  
  code UNIQUE,  
  phone,  
  email,  
  chairman_doctor_id FK,  
  chairman_start_date,  
  hospital_id FK  
)
```

```
DOCTOR(  
  doctor_id PK,  
  ssn UNIQUE,  
  first_name,  
  last_name,  
  gender,  
  birthdate,  
  phone,  
  email UNIQUE,  
  address,  
  specialization,  
  degree,  
  major_scientific_area,  
  license_number UNIQUE,  
  salary,  
  join_date,  
  department_id FK  
)
```

```
PATIENT(  
  patient_id PK,  
  patient_number UNIQUE,  
  ssn UNIQUE,  
  first_name,  
  last_name,  
  gender,  
  birthdate,  
  phone,  
  email,  
  address,  
  blood_type,  
  allergies,  
  weight_kg,  
  height_cm,  
  blood_pressure,  
  heart_rate,  
  temperature,  
  medical_history,  
  admission_date,  
  emergency_contact_name,  
  emergency_contact_phone  
)
```

```
NURSE(  
  nurse_id PK,  
  ssn UNIQUE,  
  first_name,  
  last_name,  
  gender,  
  birthdate,  
  phone,  
  email,  
  address,
```

```
role_type,  
salary,  
shift_type,  
join_date,  
department_id FK  
)
```

```
HOSPITAL(  
hospital_id PK,  
name,  
address,  
phone,  
email  
)
```

```
OPERATION_ROOM(  
room_id PK,  
department_id FK,  
hospital_id FK,  
room_number,  
room_type,  
building,  
floor_number,  
capacity,  
status  
)
```

```
APPOINTMENT(  
appointment_id PK,  
patient_id FK,  
doctor_id FK,  
scheduled_at,  
reason,  
status,
```

```
fee,  
payment_status,  
payment_date,  
cancelled_at,  
cancellation_reason,  
refund_amount,  
refund_date  
)
```

```
SURGERY(  
surgery_id PK,  
patient_id FK,  
appointment_id FK,  
lead_surgeon_dr_id FK,  
room_id FK,  
recovery_room_id FK,  
surgery_type,  
surgery_category,  
diagnosis,  
anesthesia_type,  
risk_level,  
consent_signed,  
fasting_confirmed,  
scheduled_start,  
scheduled_end,  
actual_start,  
actual_end,  
surgery_status,  
complications,  
recovery_notes,  
pain_level,  
discharge_status,  
outcome  
)
```

```
SURGICAL_TEAM(  
  team_id PK,  
  surgery_id FK,  
  doctor_id FK,  
  nurse_id FK,  
  member_type,  
  role_in_surgery,  
  assigned_at  
)
```

```
EQUIPMENT(  
  equipment_id PK,  
  room_id FK,  
  name,  
  equipment_type,  
  status,  
  purchase_date,  
  last_maintained_date  
)
```

```
SURGERY_EQUIPMENT(  
  id PK,  
  surgery_id FK,  
  equipment_id FK,  
  used_at,  
  returned_at,  
  condition_after  
)
```

```
MEDICATION(  
  medication_id PK,  
  name,  
  category,
```

```
description
)
```

```
PRESCRIPTION(  
  prescription_id PK,  
  doctor_id FK,  
  patient_id FK,  
  surgery_id FK,  
  prescription_date  
)
```

```
PRESCRIPTION_MEDICATION(  
  id PK,  
  prescription_id FK,  
  medication_id FK,  
  times_per_day,  
  dose_mg,  
  directions,  
  start_date,  
  end_date  
)
```

```
SCAN_FILE(  
  scan_id PK,  
  patient_id FK,  
  uploaded_by_doctor_id FK,  
  surgery_id FK,  
  file_path,  
  file_type,  
  scan_date,  
  description,  
  uploaded_at  
)
```

```
EXAMINATION(  
  exam_id PK,  
  doctor_id FK,  
  patient_id FK,  
  exam_date,  
  hours_per_week,  
  notes  
)
```

```
CONTACT_FORM(  
  contact_id PK,  
  status,  
  full_name,  
  email,  
  subject,  
  message,  
  submitted_at  
)
```

```
USER_ACCOUNT(  
  user_id PK,  
  is_active,  
  username UNIQUE,  
  email UNIQUE,  
  password_hash,  
  role,  
  linked_id,  
  created_at,  
  last_login  
)
```

Relationships

`DOCTOR.department_id → DEPARTMENT.department_id`

`DEPARTMENT.chairman_doctor_id → DOCTOR.doctor_id`

`NURSE.department_id → DEPARTMENT.department_id`

`OPERATION_ROOM.department_id → DEPARTMENT.department_id`

`OPERATION_ROOM.hospital_id → HOSPITAL.hospital_id`

`APPOINTMENT.patient_id → PATIENT.patient_id`

`APPOINTMENT.doctor_id → DOCTOR.doctor_id`

`SURGERY.patient_id → PATIENT.patient_id`

`SURGERY.appointment_id → APPOINTMENT.appointment_id`

`SURGERY.lead_surgeon_dr_id → DOCTOR.doctor_id`

`SURGERY.room_id → OPERATION_ROOM.room_id`

`SURGERY.room_id → OPERATION_ROOM.room_id`

`SURGERY.recovery_room_id → OPERATION_ROOM.room_id`

`SURGICAL_TEAM.surgery_id → SURGERY.surgery_id`

`SURGICAL_TEAM.doctor_id → DOCTOR.doctor_id`

`SURGICAL_TEAM.nurse_id → NURSE.nurse_id`

`EQUIPMENT.room_id → OPERATION_ROOM.room_id`

`SURGERY_EQUIPMENT.surgery_id → SURGERY.surgery_id`

`SURGERY_EQUIPMENT.equipment_id → EQUIPMENT.equipment_id`

`PRESCRIPTION.doctor_id → DOCTOR.doctor_id`

`PRESCRIPTION.patient_id → PATIENT.patient_id`

`PRESCRIPTION.surgery_id → SURGERY.surgery_id`

`PRESCRIPTION_MEDICATION.prescription_id → PRESCRIPTION.prescription_id`

`PRESCRIPTION_MEDICATION.medication_id → MEDICATION.medication_id`

`SCAN_FILE.patient_id → PATIENT.patient_id`

`SCAN_FILE.uploaded_by_doctor_id → DOCTOR.doctor_id`

`SCAN_FILE.surgery_id → SURGERY.surgery_id`

`EXAMINATION.doctor_id → DOCTOR.doctor_id`

`EXAMINATION.patient_id → PATIENT.patient_id`